DEREK CHIBUZOR

Los Angeles, CA | (201) 316-3229 | derekchibuzor777@gmail.com | linkedin.com/in/derekchibuzor | derekc22.github.io

EDUCATION

University of Southern California Los Angeles, CA

Master of Science in Mechanical Engineering Aug. 2025-Dec. 2026

* GPA: 3.90/4.00 | Robot Dynamics & Control, Linear Systems, Flight Vehicle Stability & Control, Optimization Methods

Bachelor of Science in Aerospace Engineering Aug. 2021-May 2025

* GPA: 3.62/4.00 | Computer-aided Design, Dynamic Systems, Linear Control, Computational Methods, Flight Mechanics

WORK EXPERIENCE

Neros Technologies Torrance, CA

Controls Engineer, Autonomy Jan. 2026-Present

* Develop performant **guidance, navigation, and control** algorithms for deployment on compute-constrained UAV platforms.
* Contribute to internal **6-DoF flight simulator** with custom dynamics, control, state estimation, and perception modules.
* Conduct **hardware-in-the-loop** (HIL) testing to tune controllers, perform system identification, and characterize software performance.

Dynamic Robotics and Control Laboratory Los Angeles, CA

Research Assistant, Sim2Real Jan. 2025-Present

* Engineered **system identification** framework for **Sim2Real** transfer via gradient- and sampling-based **hybrid optimization**.
* Trained **evolutionary algorithms** and **machine learning** models to improve **actuator dynamics** modelling by 78% in 24-DoF robot.
* Constructed **diffusion model** framework for generating full-body **loco-manipulation** trajectories from Unitree G1 motion capture data.

Lawrence Livermore National Laboratory Livermore, CA

Computational Engineer Intern June 2025-Aug. 2025

* Wrote IK task-space **PD controller** for 6-DoF UR3e, saving 5+ hours of manual path planning while achieving 0.1 mm tracking error.
* Trained **reinforcement learning** policies for pick and place tasks, reducing episode length by 20% with transformer feature extractor.
* Utilized implicit and explicit **structural dynamics** codes to perform **modal analysis** of jointed beam subject to broadband excitation.

Northrop Grumman Roy, UT

Mechanical Engineer Intern June 2023-Aug. 2024

* Constructed 2-DoF reduced-order Simulink model of subterranean **shock-isolated system**, saving 240 hours of FEA computation.
* Designed **CAD** assemblies, performed **FEA**, and wrote RFIs for elastomeric shock isolators and maintenance access doors.

Amazon & Information Sciences Institute Los Angeles, CA

GNC Engineer Intern June 2022-Aug. 2022

* Developed **vision-based navigation** software, enabling pose estimation for 3-DoF **rendezvous and proximity operations** (RPO).
* Achieved sub-160 ms latency localization and tracking of 12 infrared LED targets up to 2.50 m from Raspberry Pi NoIR camera.

SKILLS & CREDENTIALS

* **Programming:** Python, C++, MATLAB, Java, JavaScript
* **Software:** Simulink, Siemens NX, PyTorch, MuJoCo, CasADi, JAX, OpenCV, ROS2, LabVIEW
* **Hardware:** Raspberry Pi, Arduino, Multimeter, Oscilloscope

PROJECTS

* **6-DoF Flight Simulator**: Flight simulator written in C++. Custom dynamics, control, state estimation, etc. with FlightGear integration.
* **Multi-Agent DMPC**: Distributed and decentralized model predictive control (MPC) for high-DoF multi-agent systems.
* **Dual-Axis Control System**: Sun-seeking, multithreaded, 2-DoF electromechanical solar array articulation module for 3U CubeSat.